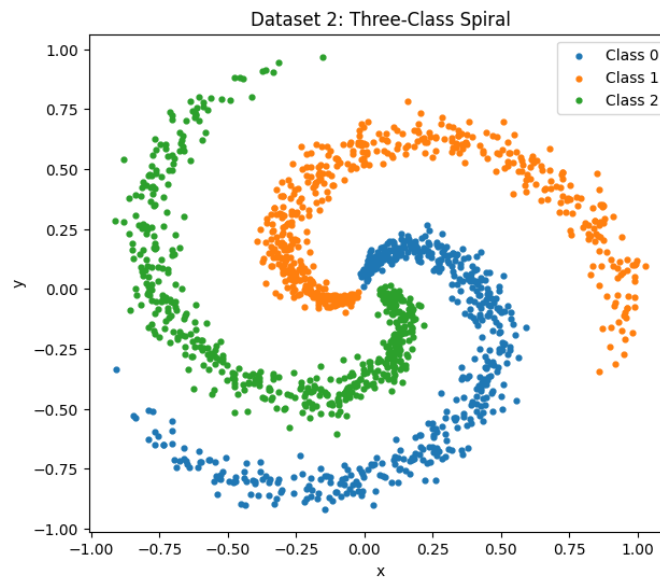
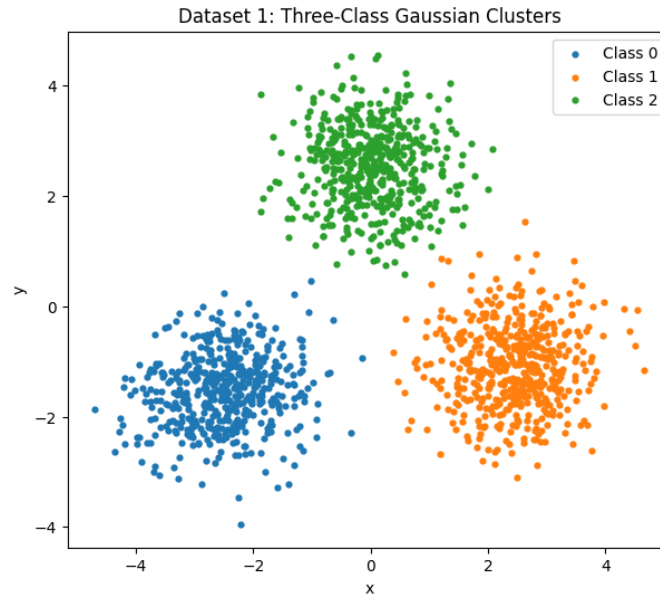


Task 1 — Multi-class Classification using FCNN

Objective and datasets

Task 1 implements a fully connected neural network (FCNN) and backpropagation from scratch using Python/NumPy for two synthetic 3-class, 2-dimensional datasets.

- Dataset 1: Three-class Gaussian clusters — 1500 samples, 3 classes and 2 input variables.
- Dataset 2: Three-class spiral — 1500 samples, 3 classes and 2 input variables; nonlinearly separable.
- Each dataset uses a 60% / 20% / 20% training-validation-test split: 900 / 300 / 300 samples.



FCNN formulation

The notebook uses sigmoid activations, one-hot targets, Xavier weight initialization, squared-error instantaneous loss, and stochastic gradient descent.

Forward propagation: $z^{(l)} = a^{(l-1)}W^{(l)} + b^{(l)}$

Activation: $a^{(l)} = \sigma(z^{(l)})$, $\sigma(z) = 1 / (1 + e^{-z})$

Sigmoid derivative: $\sigma'(z) = a(1 - a)$

Instantaneous squared-error loss: $L = \frac{1}{2} \sum_k (y_k - \hat{y}_k)^2$

Output delta: $\delta^L = (a^L - y) \odot a^L(1 - a^L)$

Hidden delta: $\delta^l = (\delta^{l+1} W^{l+1T}) \odot a^l(1 - a^l)$

SGD update: $W^l \leftarrow W^l - \eta(a^{l-1T}\delta^l)$, $b^l \leftarrow b^l - \eta\delta^l$

Accuracy: correctly classified samples / total samples

Architecture search and selection

Architecture notation is (input nodes, hidden nodes, output nodes). Dataset 1 uses one hidden layer; Dataset 2 uses two hidden layers. Selection is based on validation accuracy.

Dataset 1 — Gaussian clusters

Architecture	Train	Validation
(2, 3, 3)	100.00%	99.67%
(2, 5, 3)	100.00%	99.67%
(2, 8, 3)	99.89%	100.00%
(2, 10, 3)	100.00%	100.00%
(2, 15, 3)	99.89%	100.00%
(2, 20, 3)	99.89%	100.00%

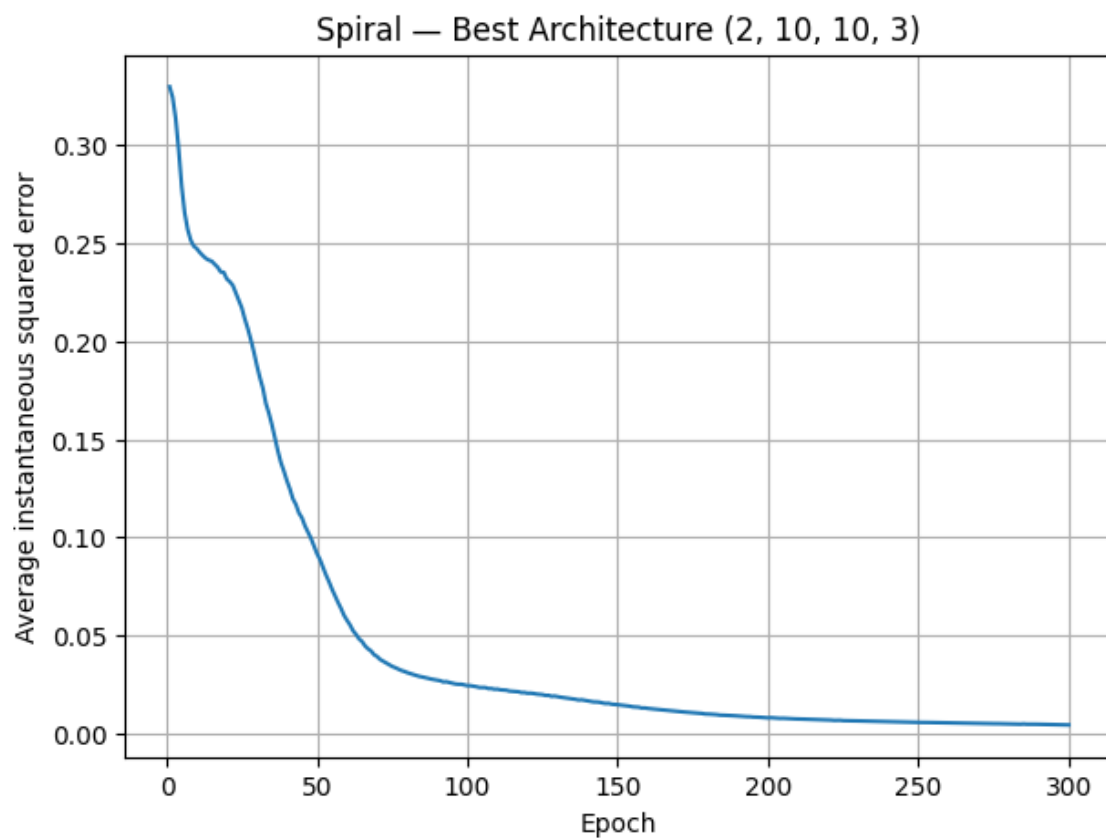
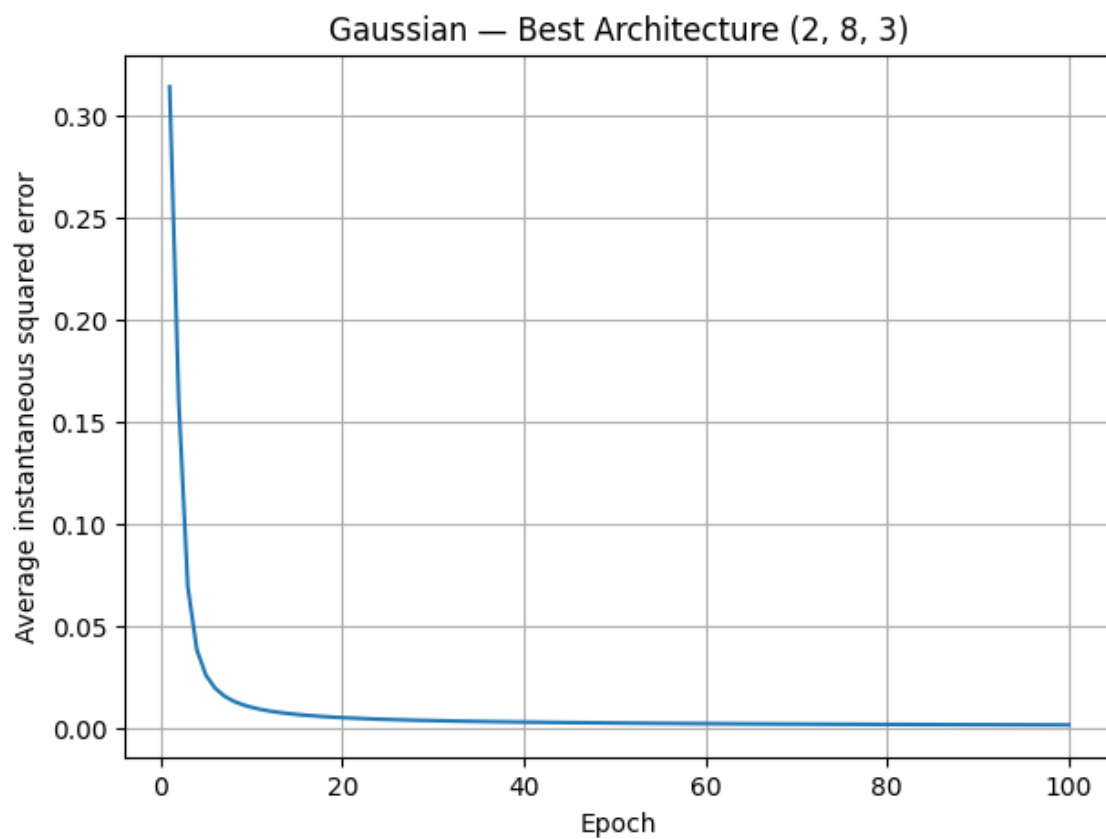
Selected architecture: (2, 8, 3).

Dataset 2 — Spiral

Architecture	Train	Validation
(2, 5, 5, 3)	99.11%	98.67%
(2, 8, 8, 3)	99.00%	97.67%
(2, 10, 10, 3)	99.78%	100.00%
(2, 15, 10, 3)	99.89%	100.00%
(2, 15, 15, 3)	99.78%	100.00%
(2, 20, 15, 3)	99.89%	100.00%
(2, 20, 20, 3)	99.89%	99.67%

Selected architecture: (2, 10, 10, 3).

Average-error plots

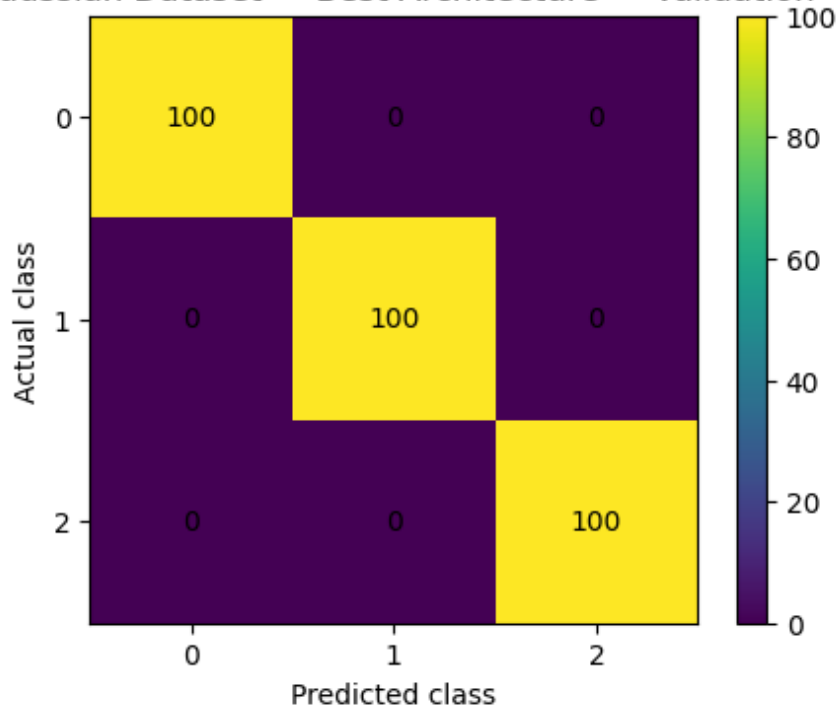


Validation and test results

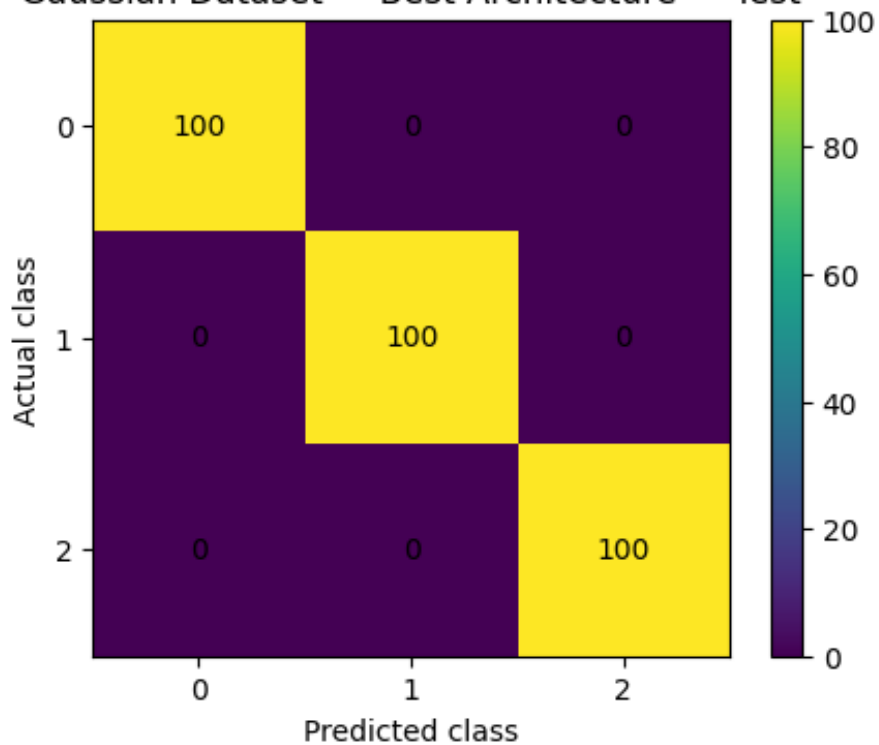
Gaussian — (2, 8, 3)

Set	Accuracy	Confusion matrix
Validation	100.00%	[[100, 0, 0], [0, 100, 0], [0, 0, 100]]
Test	100.00%	[[100, 0, 0], [0, 100, 0], [0, 0, 100]]

Gaussian Dataset — Best Architecture — Validation



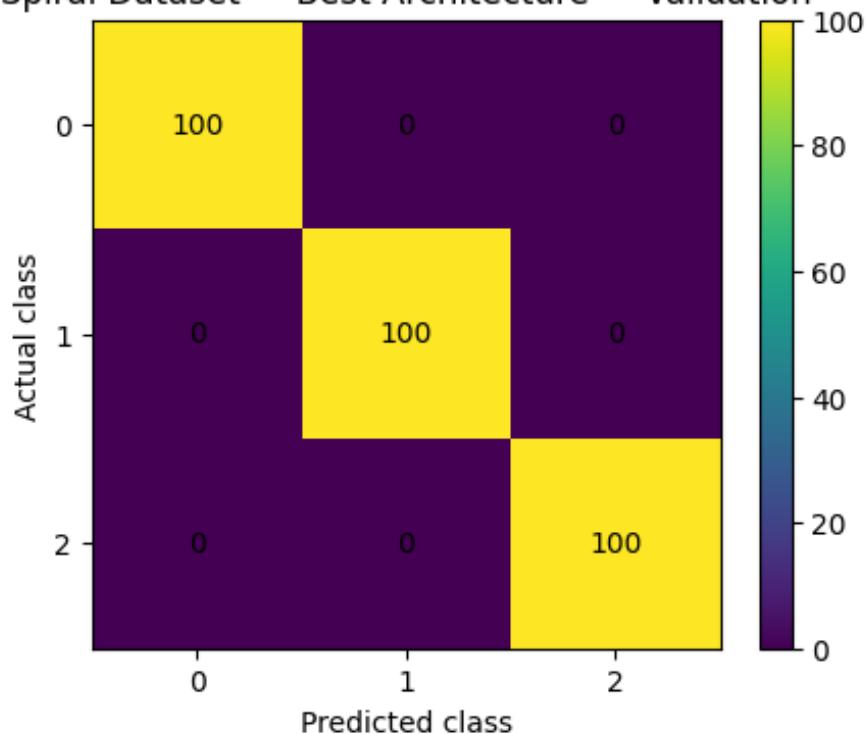
Gaussian Dataset — Best Architecture — Test



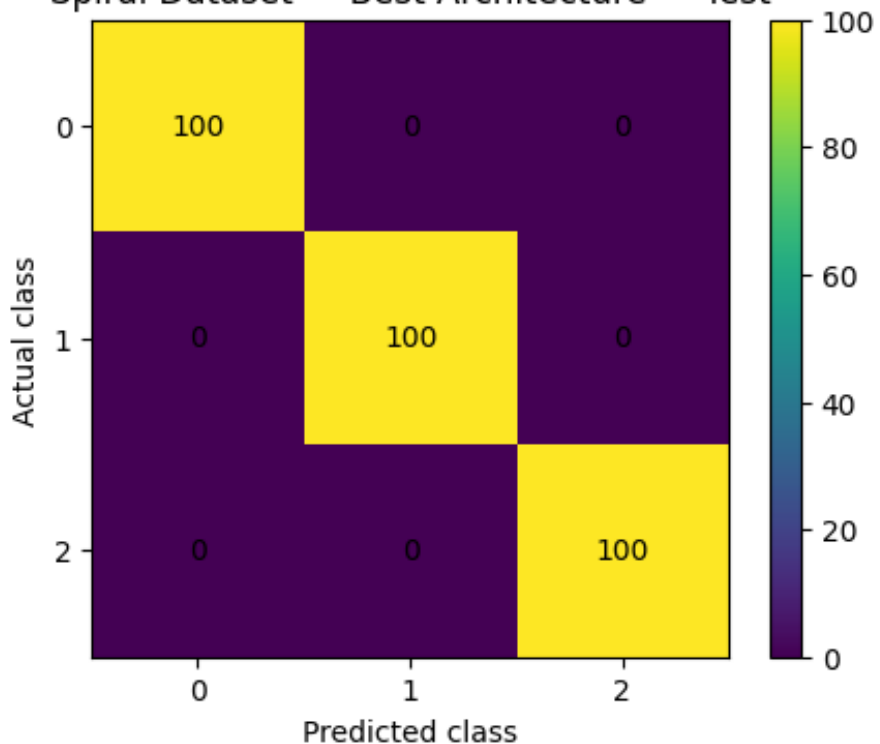
Spiral — (2, 10, 10, 3)

Set	Accuracy	Confusion matrix
Validation	100.00%	[[100, 0, 0], [0, 100, 0], [0, 0, 100]]
Test	100.00%	[[100, 0, 0], [0, 100, 0], [0, 0, 100]]

Spiral Dataset — Best Architecture — Validation

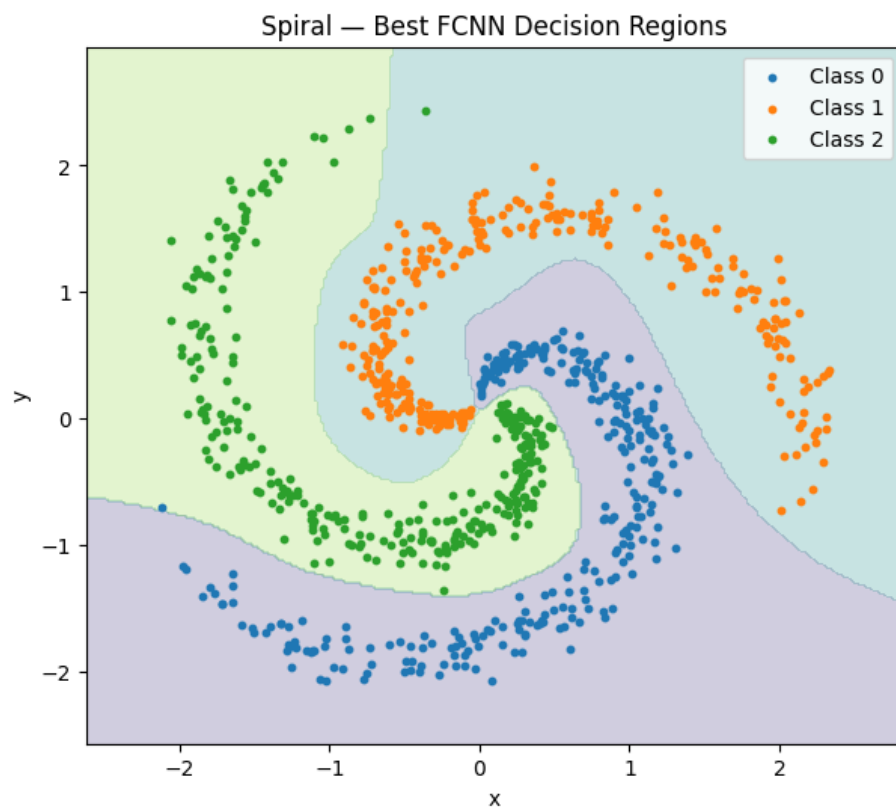
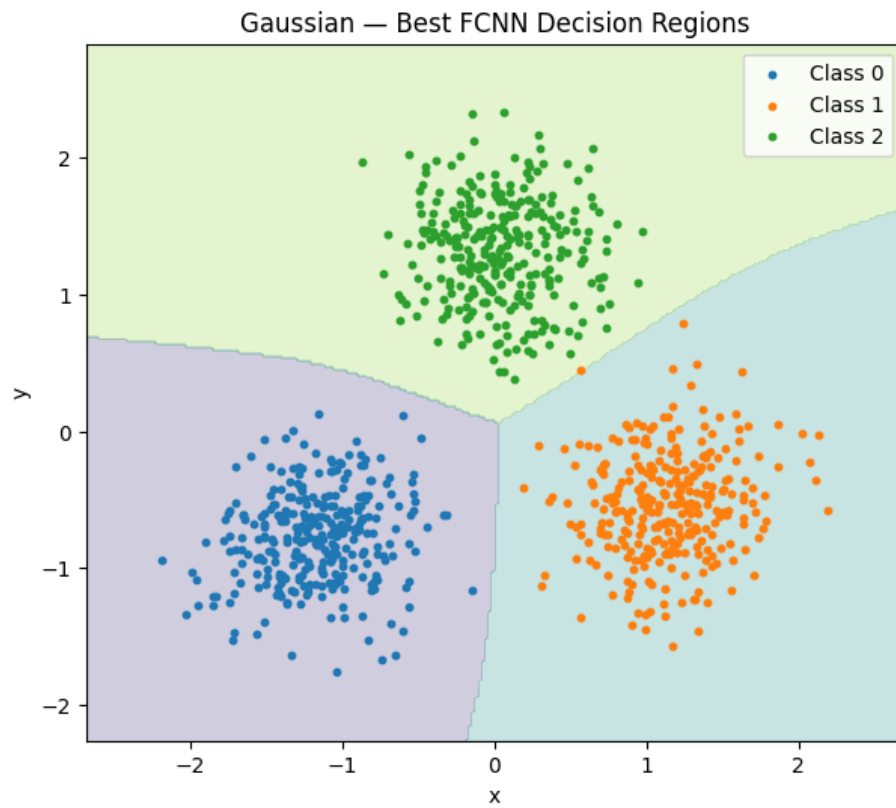


Spiral Dataset — Best Architecture — Test



Decision regions

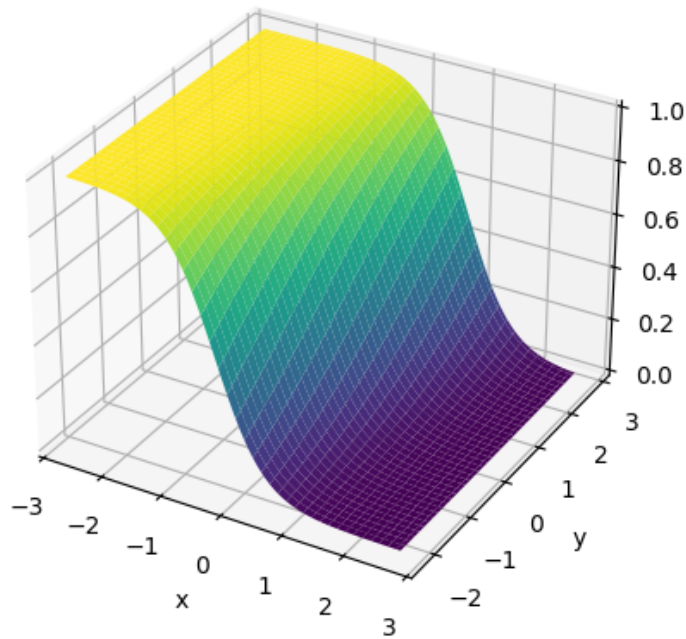
We evaluate the trained network on a dense 2-D grid and superimpose the training samples.



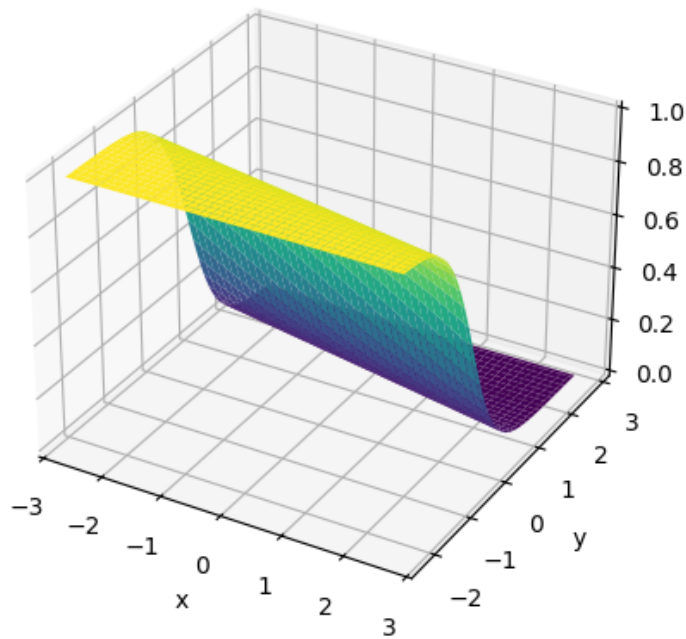
Hidden/output node surfaces

We generate $z = f(x,y)$ surfaces for hidden and output nodes for the requested training, validation and test splits.

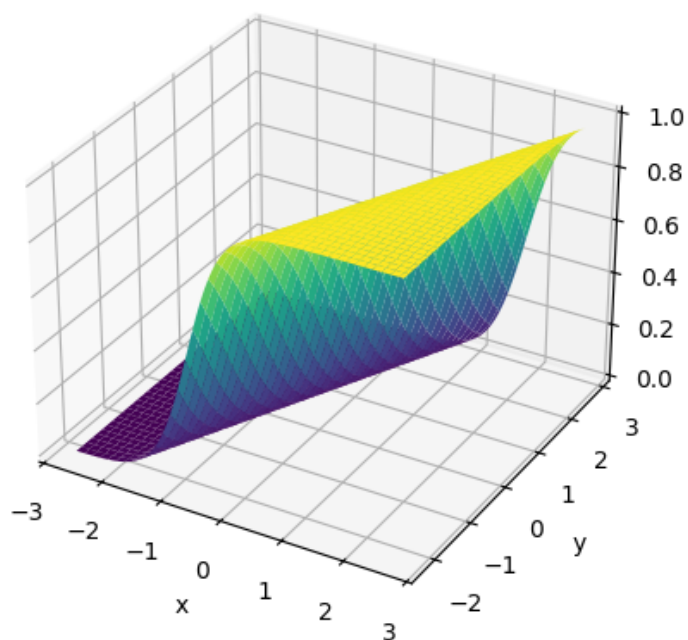
Gaussian — X_train | Layer 1, Node 1



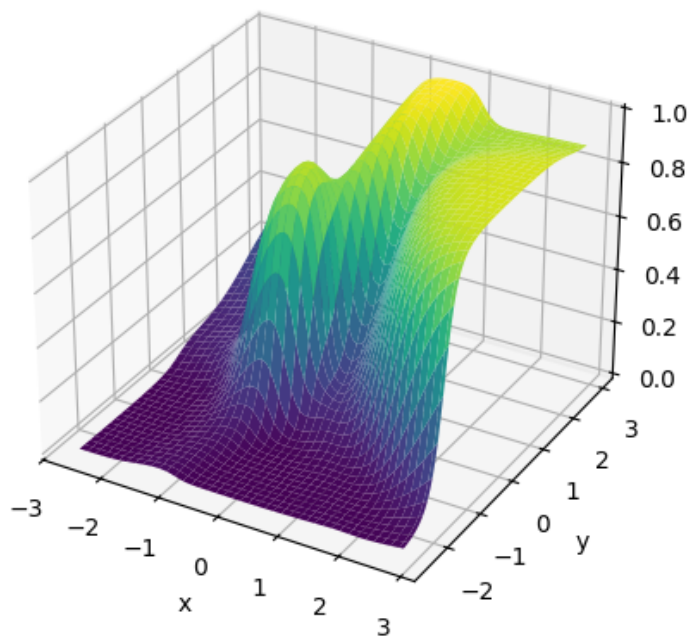
Gaussian — X_train | Layer 1, Node 2



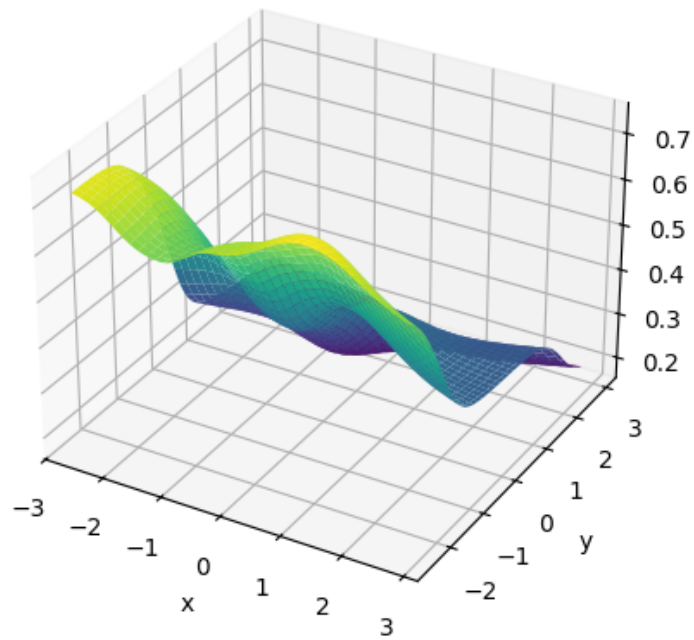
Gaussian — X_train | Layer 1, Node 3



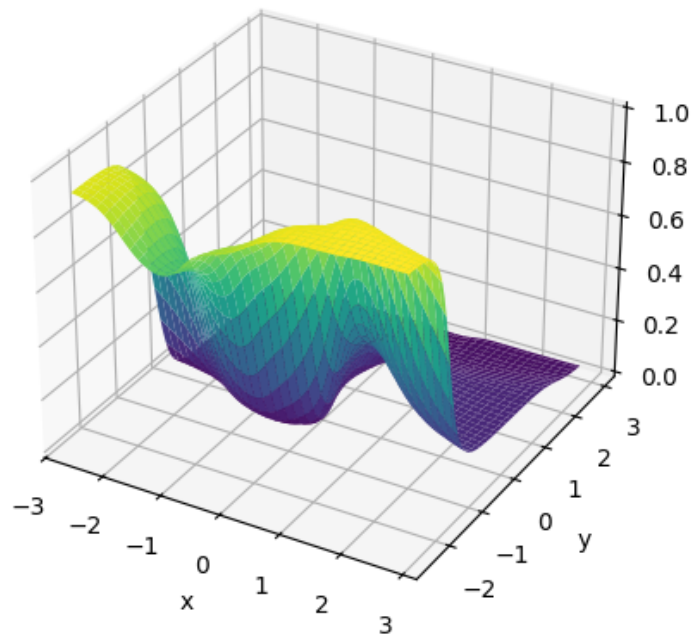
Spiral — X_train | Layer 2, Node 8



Spiral — X_train | Layer 2, Node 9



Spiral — X_train | Layer 2, Node 10



Single-neuron / linear-model comparison

Dataset	Model	Train	Test
Gaussian	FCNN	99.89%	100.00%
Gaussian	Single	99.89%	99.67%
Spiral	FCNN	99.78%	100.00%
Spiral	Single	51.89%	46.00%

The results show a large performance difference on the spiral dataset between the FCNN and the single-neuron/linear model.

Inferences from the results

- Several Gaussian architectures reach 100% validation accuracy; the notebook selects (2, 8, 3).
- The selected Spiral architecture is (2, 10, 10, 3), with 100% validation accuracy.
- Both selected architectures obtain 100% test accuracy in the notebook.
- The single-neuron/linear model performs poorly on the spiral dataset compared with the FCNN.